

PHILOSOPHY & APPLIED LOGIC

Official Grade 12 Textbook - Humanities Division

ACADEMIC YEAR: 2025 / 2026 | GENERAL SECONDARY GRADE 12 (HUMANITIES & SOCIAL SCIENCES)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

The comprehensive official Egyptian Ministry of Education curriculum for Grade 12 Philosophy and Applied Logic, covering environmental ethics, bioethics, professional values, inductive scientific reasoning, formal truth tables, and AI reasoning.

CORE CURRICULUM HIGHLIGHTS:

- Environmental ethics: Singer, Regan, Hans Jonas, Aldo Leopold, Arne Naess deep ecology
- Bioethics: In-vitro fertilization, surrogate motherhood, genetic screening, cloning, informed consent elements
- John Stuart Mill five inductive methods and Francis Bacon idols of the mind
- Truth tables for negation, conjunction, disjunction, conditional and biconditional relations
- Lotfi Zadeh fuzzy logic and non-monotonic AI reasoning systems

APPROVED CHAPTER MODULES:

- Ch 1: Philosophy & Environmental Issues	pp. 1-28 (12 core units)
- Ch 2: Philosophy & Bioethics (Medical Dilemmas)	pp. 29-58 (14 core units)
- Ch 3: Philosophy of Professional Ethics	pp. 59-86 (12 core units)
- Ch 4: Values, Philosophizing & Human Freedom	pp. 87-112 (12 core units)
- Ch 5: Inductive Inference in Natural Science	pp. 113-140 (14 core units)
- Ch 6: Deductive Logic & Symbolic Systems	pp. 141-168 (15 core units)
- Ch 7: The Contemporary Hypothetico-Deductive Method	pp. 169-194 (12 core units)
- Ch 8: Cybernetics, Logic & Artificial Intelligence	pp. 195-220 (14 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Grade 12 Philosophy & Applied Logic Ministry Compendium (42p - 3.8 MB)

Full Ministerial Textbook: 220 Pages - 26.4 MB (Direct on MOE Portal)

Publisher: Ministry of Education & Technical Education - Arab Republic of Egypt

Official Portal: <https://moe.gov.eg/ar/elearning/sec3/philosophy/> - Code: MOE-PHIL-G12

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MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

The comprehensive official Egyptian Ministry of Education curriculum for Grade 12 Philosophy and Applied Logic, covering environmental ethics, bioethics, professional values, inductive scientific reasoning, formal truth tables, and AI reasoning. Environmental ethics: Singer, Regan, Hans Jonas, Aldo Leopold, Arne Naess deep ecology Bioethics: In-vitro fertilization, surrogate motherhood, genetic screening, cloning, informed consent elements John Stuart Mill five inductive methods and Francis Bacon idols of the mind Truth tables for negation, conjunction, disjunction, conditional and biconditional relations Lotfi Zadeh fuzzy logic and non-monotonic AI reasoning systems

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

40% Knowledge & Recall Definitions, laws & formulas	35% Understanding & Application Calculations, proofs & models	25% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal: <https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Ch 1: Philosophy & Environmental Issues	4 wks	8 pts	12.5%	12 core instructional units covering national standards and
Ch 2: Philosophy & Bioethics (Medical Dilemmas)	4 wks	8 pts	12.5%	14 core instructional units covering national standards and
Ch 3: Philosophy of Professional Ethics	4 wks	8 pts	12.5%	12 core instructional units covering national standards and
Ch 4: Values, Philosophizing & Human Freedom	4 wks	8 pts	12.5%	12 core instructional units covering national standards and
Ch 5: Inductive Inference in Natural Science	4 wks	8 pts	12.5%	14 core instructional units covering national standards and
Ch 6: Deductive Logic & Symbolic Systems	4 wks	8 pts	12.5%	15 core instructional units covering national standards and
Ch 7: The Contemporary Hypothetico-Deductive Method	4 wks	8 pts	12.5%	12 core instructional units covering national standards and
Ch 8: Cybernetics, Logic & Artificial Intelligence	4 wks	5 pts	12.5%	14 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: CH 1: PHILOSOPHY & ENVIRONMENTAL ISSUES

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Ch 1: Philosophy & Environmental Issues.
- [+] Synthesize and apply analytical frameworks across all 12 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Ch 1: Philosophy & Environmental Issues.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: CH 2: PHILOSOPHY & BIOETHICS (MEDICAL DILEMMAS)

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Ch 2: Philosophy & Bioethics (Medical Dilemmas).
- [+] Synthesize and apply analytical frameworks across all 14 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Ch 2: Philosophy & Bioethics (Medical Dilemmas).

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: CH 3: PHILOSOPHY OF PROFESSIONAL ETHICS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Ch 3: Philosophy of Professional Ethics.
- [+] Synthesize and apply analytical frameworks across all 12 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Ch 3: Philosophy of Professional Ethics.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

PHILOSOPHY & APPLIED LOGIC - CORE PRINCIPLES & AXIOMS	
Fundamental Identity:	$\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$
Equilibrium Balance Criterion:	$\sum F_{\text{net}} = 0, \sum M_{\text{pivot}} = 0$
Continuity & Conservation Standard:	$\text{div}(\mathbf{J}) + \partial \rho / \partial t = 0$
Normalized Standard Score:	$Z = (X - \mu) / \sigma$
PROCEDURAL ASSESSMENT & TRANSFORMATION RULES	
Linear Operator Transformation:	$T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Harmonic Synthesis Mean:	$H = n / (\sum_{i=1}^n 1/x_i)$
Sample Variance Boundary:	$s^2 = (1 / (n - 1)) \cdot \sum (x_i - \bar{x})^2$
Curricular Efficiency Ratio:	$\eta = (\text{Output}_{\text{actual}} / \text{Input}_{\text{standard}}) \cdot 100\%$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

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